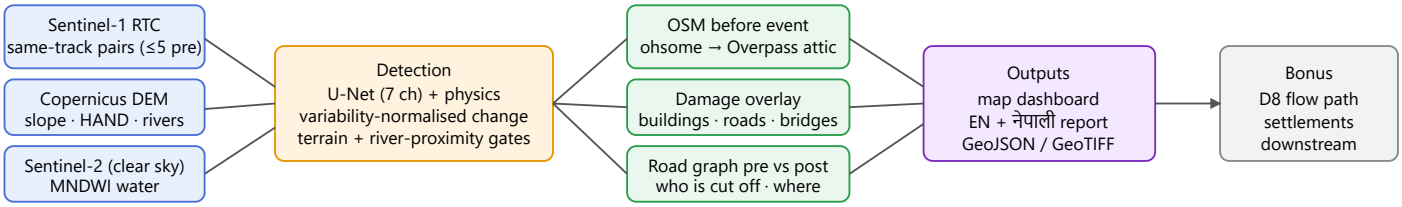
# BeaconSpace — mapping flood damage from space

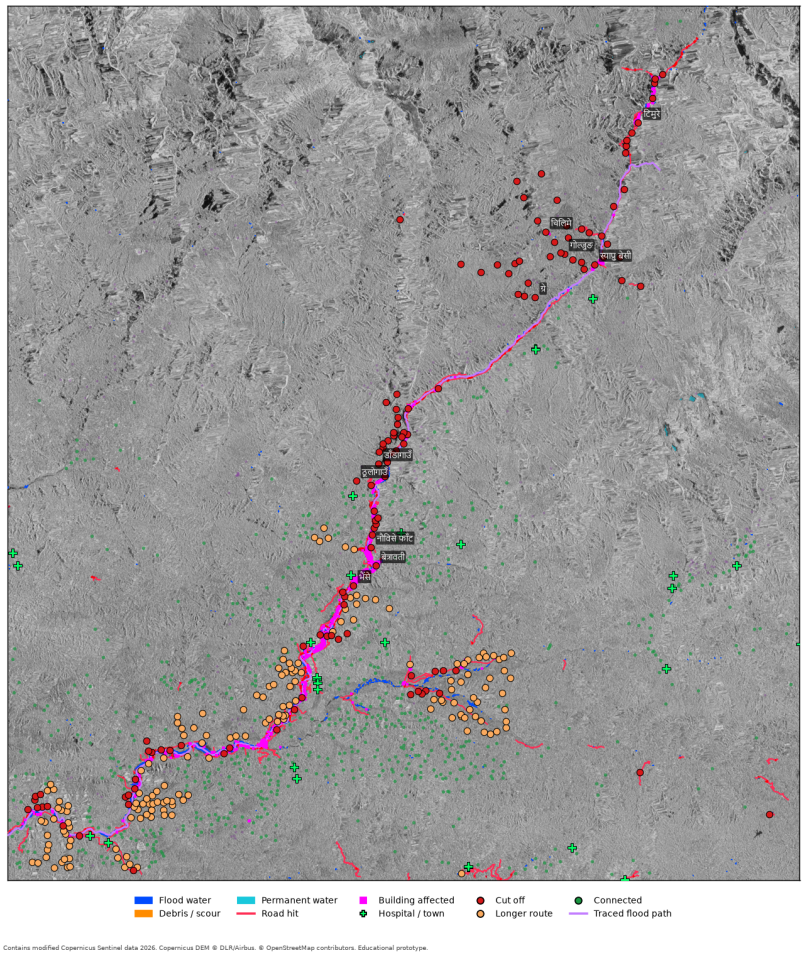
Case study: Trishuli / Bhote Koshi, 26 Aug 2026 · Multimodal AI Hackathon 2026 · Track B · AI component: flood-segmentation model · Educational prototype, not an operational tool.

## 1 · What the system does

Given a bounding box and a flood date, the system downloads Sentinel-1/2, the Copernicus DEM and a pre-event OpenStreetMap snapshot, and answers three questions: **where the flood hit** (flood water and debris/scour), **what was damaged** (buildings, roads, bridges) and **who is cut off** (settlements that lost their road route to the nearest hospital or town). A bonus module traces the flood path downstream from any upstream point. Outputs: interactive dashboard, one-page bilingual (English/Nepali) situation report, GeoTIFF/GeoJSON and a provenance file. It runs from raw data for any area/date: `python -m floodsys run --bbox W S E N --event DATE`.




**Case-study result** (3 334 km², 20 m grid, RO85 (2026-08-16 → 2026-08-28, 5 pre-event images); RO19 (2026-08-24 → 2026-09-05, 5 pre-event images); Sentinel-2 cloud-free overlap 7%): **1,364 ha** flooded or debris-covered (61 ha water, 1,303 ha debris/scour), **2,610** of 74,869 mapped buildings affected (1,984 likely destroyed), **67 km** of road and **33/151** bridges hit, **126** settlements cut off and 164 with a much longer route (7,641 buildings in cut-off settlements).




Case-study map (final run): flood/debris along the Bhote Koshi–Trishuli corridor, hit roads, affected buildings, and settlements by road access (red = cut off). Contains modified Copernicus Sentinel data 2026; © OpenStreetMap contributors.

## 2 · Data, rules and attribution

| Input | Source / use | Rule compliance |
|---|---|---|
| Sentinel-1 RTC $\gamma^0$ 10 m | Planetary Computer (from Copernicus GRD); same relative-orbit pre/post pairs, up to 5 pre-event images | allowed input |
| Sentinel-2 L2A | AWS Earth Search COGs; cloud/snow-masked by SCL; MNDWI water evidence | allowed input |
| Copernicus DEM GLO-30 | slope, HAND, D8 flow, drainage network | allowed input |
| OpenStreetMap | snapshot **2026-07-27** (before the event) via overpass-attic; the public ohsome *geometry* endpoint answered HTTP 403 from our network, so date-pinned Overpass | pre-event only; no post-event edits (in one dense tile 15 % of buildings had been edited after the snapshot, which |

| | | |
|---|---|---|
| | "attic" queries were used | shows why live data is not an option) |
| Kuro Siwo (GRD) | training the U-Net — Bountos et al., NeurIPS 2024, CC BY | listed training dataset |
| EMS EMSR927 | checking only (`scripts/evaluate_emsr927.py`); never read by the pipeline; `provenance.json` records `forbidden_inputs_used: []` | credit: "European Union, Copernicus Emergency Management Service data" |

Contains modified Copernicus Sentinel data 2026. Produced using Copernicus WorldDEM-30 © DLR e.V. 2010–2014 and © Airbus Defence and Space GmbH 2014–2018 provided under COPERNICUS by the European Union and ESA; all rights reserved. © OpenStreetMap contributors.

## 3 · Method

### 3.1 Radar pairs and change detection

**Pairing.** Acquisitions are grouped by relative orbit; a pair is a post-event pass plus up to five earlier passes on the *same* track (tracks are never compared pixel-by-pixel), preferring full AOI coverage, a short delay after the flood and a pre-image ≈12 days before. **Radiometry.** RTC $\gamma^0$ → dB, 3×3 Lee filter at 10 m (as in Kuro Siwo), linear-domain window means at 20 m. **Water** must be dark (VV < −17 dB or VH < −24 dB) *and* have dropped ≥3 dB against the pre-event mean, which removes radar shadow (dark in both images); slope ≤12° and HAND ≤40 m gates remove high ground. **Debris/scour** (mud, boulders, scoured channel) raises or changes backscatter in both polarisations. We use a speckle-robust log-ratio, normalise it by *each pixel's own natural variability* (temporal std over the pre-event series), apply hysteresis thresholds (2.5σ / 4σ, ≥2.5 dB) derived from the empirical noise of stable hillslopes, and keep only changes within 150 m of the drainage network (catchment ≥5 km²). **Optical** evidence only *adds* flood water where the sky is clear; it never vetoes radar. **Multi-track**: union of tracks with an agreement count as confidence (523 ha seen by both tracks).

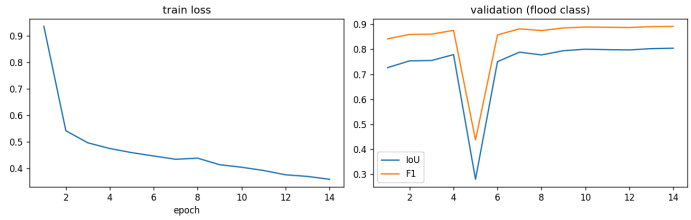
> **A mistake we found and fixed (null-event test).** Our first detector flagged 670–2 620 ha of "debris" on image pairs where *nothing happened* — rice terraces and monsoon-wet soils change backscatter by several dB. Two changes, chosen using only quiet-period pairs and physical reasoning (not EMS data), cut late-season false alarms by roughly 5–10×: per-pixel variability from the pre-event series, and the river-proximity rule. Parameters were frozen before the EMS comparison below.

### 3.2 Damage and cut-off analysis

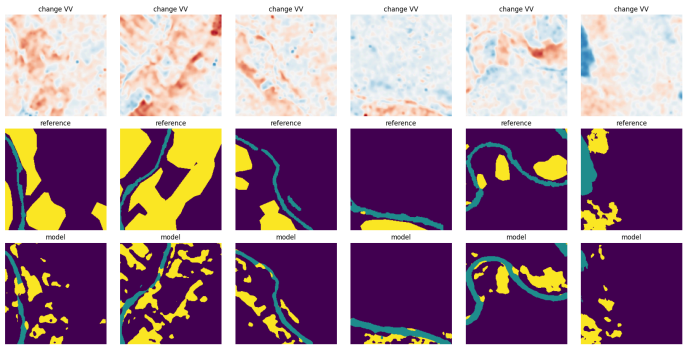
**Buildings** (OSM footprints): *likely destroyed* ≥50 % inside the mask; *affected* touching it (±10 m); *exposed* within 40 m. **Roads/bridges**: ≥10 m inside the mask (buffered half a pixel); bridges = ways tagged `bridge`. **Who is cut off**: a vertex-level road graph (2,945 segments blocked in the final run); a multi-source Dijkstra from 883 hospitals/clinics/towns gives each settlement its nearest destination before and after. *cut_off* = route exists before but not after; *longer_route* = route ≥1.5× and >1 km longer. The first blocked segment on the old route is reported as the place to clear. Settlements = OSM places plus building clusters (DBSCAN) named by the nearest place. **Flow path**: D8 flow directions from the DEM, followed from the snapped upstream cell to the area edge (90.11 km, 54 settlements in the corridor within 30 m of height above the path).

## 4 · AI component: flood-segmentation model

**Model.** U-Net with a ResNet-34 encoder (ImageNet-initialised), 7 input channels (post VV/VH, pre VV/VH, smoothed log-ratio VV/VH, terrain slope), 3 classes (no water / permanent water / *new* flood water), exported to ONNX (98 MB, CPU inference; a unit test proves training and live feature construction are identical). **Training** on Colab T4: 11700 patches from 5 train shards; epochs=14; cross-entropy with class weights + Dice, AdamW + one-cycle LR, ±1.5 dB gain and random 2× down-scale augmentation (the system runs at 20 m). Data: Kuro Siwo GRD (Bountos et al., NeurIPS 2024). **In the pipeline** the network proposes flood water and the physics/terrain gates still apply.




Training loss and validation flood IoU (random 8 % of training patches — optimistic, not the headline). The dip at epoch 5 is a one-off instability; the best checkpoint was kept.




Held-out examples: change-VV, reference, model (yellow = flood, teal = permanent water).

| Evaluation (flood class) | IoU | F1 | Precision | Recall | n |
|---|---|---|---|---|---|
| Kuro Siwo test shards, 10 m | 0.72 | 0.84 | 0.77 | 0.92 | 14,400 patches |
| … at 20 m (live-system resolution) | 0.71 | 0.83 | 0.75 | 0.93 | same |
| **Events never seen in training** (6 events, 3,600 patches): patch-weighted mean of per-event IoU, 10 m / 20 m | **0.46 / 0.46** (per-event range 0.15–0.92, median 0.37 at 10 m) | | | | 6 events |
| **Mountain subset** (mean slope ≥10°), 10 m / 20 m | 0.35 / 0.30 | 0.52 / 0.46 | 0.45 / 0.37 | 0.62 / 0.62 | 805 patches |
| Physics threshold baseline vs model, same subset, 10 m | 0.50 vs 0.73 | 0.67 vs 0.84 | 0.81 vs 0.77 | 0.57 vs 0.93 | 600 patches |
| False flood pixels on flood-free held-out patches (20 m) | 2.6% of land | | | | ≤800 patches |

| Unseen event | patches | IoU 10 m / 20 m | P / R (10 m) |
|---|---|---|---|
| event 118 | 342 | 0.15 / 0.18 | 0.27 / 0.25 |
| event 130 | 373 | 0.92 / 0.93 | 0.95 / 0.97 |
| event 147 | 485 | 0.25 / 0.27 | 0.28 / 0.65 |
| event 277 | 873 | 0.74 / 0.74 | 0.79 / 0.93 |
| event 279 | 327 | 0.47 / 0.41 | 0.58 / 0.72 |
| event 427 | 1200 | 0.27 / 0.28 | 0.38 / 0.48 |

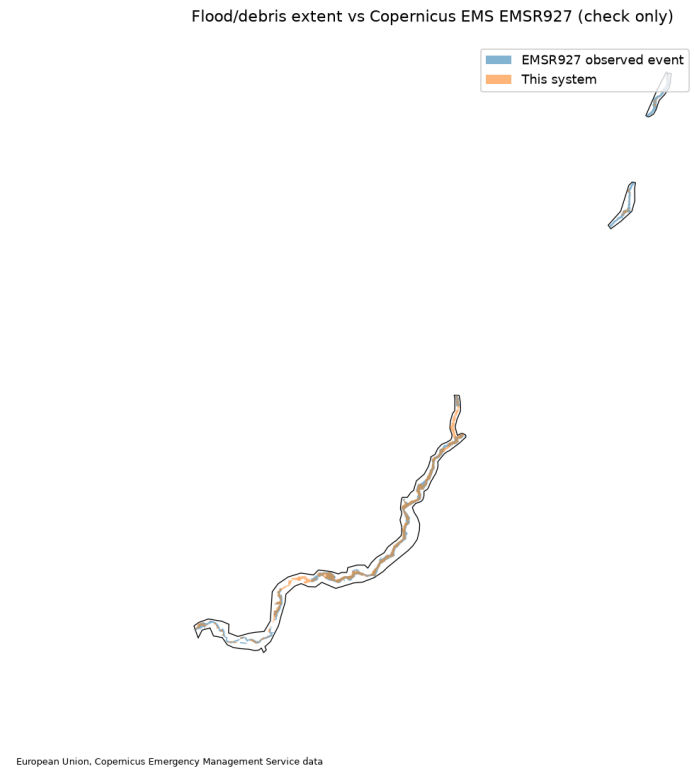
**Honest reading.** The all-patch numbers above are **not** an event-held-out test: 19 of the 25 events in the test shards also occur in our training patches (Hugging Face's split is not by event), and the validation curve is a random patch split. The clean test is the 6 events absent from training: IoU 0.15–0.92, **patch-weighted mean 0.46** — the model generalises well to some places and poorly to others. It also beats the threshold baseline on the same patches, but is weak on steep terrain (IoU 0.35, precision 0.45) and over-predicts flood on dry land (2.6%), which is why its output is gated by slope, HAND and permanent-water masks. Kuro Siwo has no debris-flow labels, so debris comes from the change detector. In the Trishuli case study the model changes the result only slightly (see ablation below).

## 5 · Case study vs Copernicus EMS EMSR927 (check only)

EMS mapped four areas (Syapru Besi, Timure, Bidur, Phosretar; 5,132 ha) as one class, “mass movement”, 1,309 ha. We compare our flood+debris footprint, OSM buildings and roads inside these areas. The study area was widened west to cover the Phosretar area (a coverage choice only). Detector parameters were set before looking at EMS; EMS was read twice (before/after the false-alarm fix) and nothing was tuned on it.

| Inside EMS areas | v1 first detector | final system |
|---|---|---|
| Our extent / EMS extent (ha) | 1,262 / 1,309 | 981 / 1,309 |
| IoU | 0.50 | 0.48 |
| Precision / recall | 0.68 / 0.65 | 0.76 / 0.57 |
| … with 30 m tolerance | 0.76 / 0.81 | 0.82 / 0.76 |
| EMS affected buildings found (≤20 m) | 2,557 / 3,816 (67%) | 2,153 / 3,816 (56%) |
| Road: EMS affected km covered / our precision | 88% / 45% | 84% / 49% |

Road precision is low because OSM maps many tracks that EMS did not assess. EMS counts only affected buildings, so building precision cannot be computed.




Final extent (orange) vs EMS observed event (blue) inside the EMS areas. European Union, Copernicus Emergency Management Service data.

**Reading.** We trace the same corridor, but the final detector trades recall for precision: EMS "mass movement" also includes hill-slope landslide scars away from rivers, which our river-proximity rule deliberately does not map (and radar change there is indistinguishable from vegetation/soil change). Using the whole EMS-area comparison, the system finds 57% of EMS's event area and 56% of its affected buildings.

**What does the AI model add here? (ablation)** Running the same case study *without* the network: flood+debris 1,351 ha vs 1,364 ha with it; EMS-area IoU 0.48 vs 0.48; EMS-affected buildings found 56% vs 56%; road coverage 84% vs 84%. **The network changes the Trishuli result by about 1 %**: the dominant class (debris/scour) comes from the change detector, EMS has no separate flood-water class that could reward the network, and the model's flood water (61 ha vs 57 ha) is small and still gated by terrain. The network's value is demonstrated on Kuro Siwo (open-water floods, §4), not on this debris-flow case.

## 6 · False alarms: the null-event test

For fake "events" before the real flood we run the identical pipeline on same-track pairs where nothing happened; everything flagged is a false alarm by construction (area 3333.9 km², ha flagged).

| Detector | Pre → post pair | false flood ha | false debris ha | ha per 1000 km² |
|---|---|---|---|---|
| v1: 2 pre images | 2026-06-25 → 2026-07-07 | 11 | 1541 | 466 |
| v1: 2 pre images | 2026-07-07 → 2026-07-19 | 23 | 2620 | 793 |
| v1: 2 pre images | 2026-07-19 → 2026-07-31 | 6 | 746 | 226 |
| v1: 2 pre images | 2026-07-31 → 2026-08-12 | 4 | 668 | 202 |
| final, physics | 2026-06-25 → 2026-07-07 | 23 | 460 | 145 |
| final, physics | 2026-07-07 → 2026-07-19 | 39 | 760 | 239 |
| final, physics | 2026-07-19 → 2026-07-31 | 15 | 55 | 21 |
| final, physics | 2026-07-31 → 2026-08-12 | 9 | 42 | 15 |
| final, physics | REAL EVENT 2026-08-16 → 2026-08-28 | 58 | 823 | – |
| final, physics | REAL EVENT 2026-08-24 → 2026-09-05 | 32 | 953 | – |
| final, + AI model | 2026-06-25 → 2026-07-07 | 29 | 466 | 149 |
| final, + AI model | 2026-07-07 → 2026-07-19 | 55 | 763 | 246 |
| final, + AI model | 2026-07-19 → 2026-07-31 | 0 | 55 | 16 |
| final, + AI model | 2026-07-31 → 2026-08-12 | 1 | 41 | 12 |
| final, + AI model | REAL EVENT 2026-08-16 → 2026-08-28 | 59 | 837 | – |
| final, + AI model | REAL EVENT 2026-08-24 → 2026-09-05 | 31 | 958 | – |

Late-season pairs (pre-event series in the monsoon): mean false alarm **712 ha** with the first detector → **61 ha** final (physics) → 48 ha with the AI model, against ≈1 000–1 200 ha of debris flagged for the real event. Early-July pairs stay noisy because their pre-event series ends before the monsoon began — **a real limitation**: the detector needs pre-event images from the same season.

**Second Himalayan event, no reference map (robustness only).** The unchanged system was run on the June 2021 Melamchi flood (small area, different year, Sentinel-1A/1B, a 2021 OSM snapshot chosen automatically): it completed end to end and mapped 156 ha (36 ha water, 119 ha debris), 219 affected buildings of 17,353, 7/34 bridges hit and 31 cut-off settlements; the traced flood path (27.14 km) follows the detected corridor

through Melamchi Bazaar. **We have no reference for this event, so this shows the system runs on a new place and date — it is not an accuracy claim.** Isolated patches on a side tributary look like possible false alarms.

## 7 · Limitations — what the system cannot do

- **No warning.** Sentinel satellites revisit every few days (12 days per track); the system describes the scene at the last pass, not "now", and cannot warn of a glacier collapse minutes ahead. Flood extent may have changed or receded before the post-event image.
- **Resolution.** 10–20 m pixels: single houses, small footbridges and narrow tracks are inferred from overlap with the footprint, not seen. Building status is an *estimate* (no VHR optical imagery).
- **Debris is "change", not a class.** It cannot say mud vs. boulders vs. landslide, depth or volume; hill-slope landslides away from rivers are out of scope; farmland near rivers can still false-alarm (see §6).
- **Monsoon cloud.** Sentinel-2 was usable on only 7% of the area; radar is blind to nothing about cloud but is confused by wet snow, layover/shadow in gorges, and changes in rice/crop cover.
- **Road network.** Cut-off logic only knows *mapped* roads (OSM at 2026-07-27): unmapped tracks, footpaths, helicopter or boat access and repairs since are unknown; routes leaving the analysed area are assumed intact; "settlement" = OSM place/building cluster, with no population data (not an allowed input).
- **AI model.** Trained on mostly low-relief floods; on events never seen in training its IoU ranges 0.15–0.92 (§4); weak on steep terrain; 2.6 % false-alarm rate on flood-free patches; no debris labels.
- **OpenStreetMap access.** The public ohsome geometry endpoint was blocked from our network; the Overpass fallback is slow (our 3 300 km² case took an hour or more once; cached afterwards).
- **Nepali text** is template-based and needs review by a native speaker; **no values are generated by a language model**.
- **Uncertainty is not propagated** into the counts; treat them as ± tens of percent (see EMS comparison).

## 8 · Reproducibility, responsibility, citations

Code, tests (`pytest`), Colab notebook, README and per-run `provenance.json` are in the repository; the case study runs in ≈10 min once OSM tiles are cached. This disaster is recent: no images of victims are used; outputs are labelled as educational prototypes.

**Cite:** Bountos, N.I., Sdraka, M., Zavras, A., Karavias, A., Karasante, I., Herekakis, T., Thanasou, A., Michail, D., Papoutsis, I. (2024) *Kuro Siwo: 33 billion m² under the water*, NeurIPS Datasets & Benchmarks. · Copernicus Sentinel-1/2 data 2026; Copernicus DEM GLO-30; © OpenStreetMap contributors (ODbL); European Union, Copernicus Emergency Management Service data (activation EMSR927, checking only).